

One Bank Ltd

SCO – 2022

- 1) In an election between two candidates, one got 55% of the total valid votes. 20% of the votes were invalid. If the total number of votes was 7500, what was the number of valid votes that the other candidate got?

Solution: One Candidate got = 55% of 80 = 44% votes

Other got = 80% – 44% = 36% votes

Let, total number of votes = 100x

According to the question $100x = 7500$ Or, $36x = 7500 \times 36 = 2700$

Answer: 2700

- 2) A, B and C enter into a partnership in the ratio (7/2): (4/3): (6/5). After 4 months, A increases his share 50%. If the total profit at the end of one year be Tk. 21,600, then B's share in the profit is:

Solution: Investment ratio of A, B, C = (7/2) : (4/3) : (6/5) = 105 : 40: 36

Let, investment of A, B, C be = 105x, 40x, 36x

After increasing 50% A's share,

Profit Ratio, A = $105x \times 4 + (105x \times 150 / 100) \times 8 = 1680x$

B = $40x \times 12$

C = $36x \times 12$

A: B: C = $1680x : 40x \times 12 : 36x \times 12 = 35 : 10 : 9$

Now, 54 parts = 21600 tk

10 parts = $(21600 \times 100 / 54) = 4000$ taka

Answer: B's share profit = 4000 taka.

Premier Bank Ltd

Trainee Junior Officer – 2021

- 1) The ratio of the salary of A,B & C is 7:5:3.if B gets tk 222 more than what C gets, then what is the salary of A?

Solution

Salary of A, B, C is = 7x, 5x, 3x [x is the common term]

Now, $5x - 3x = 222$ Or, $x = 111$

The salary of A is = $7 \times 111 = 777$

Answer: Tk 777

- 2) The average age of a committee of eight members is 40 years. A member aged 55 years retired and his place was taken by another member aged 39 years. The average age of the present committee is?

Solution

Difference in age of last person = $55 - 39 = 16$

This change will be distributed among all 8 members,
hence the average of group reduced by $16 \div 8 = 2$ years

So average age of present committee = $40 - 2 = 38$ years Answer: 38 years

Social Islami Bank Ltd
Probationary Officer - 2021

- 1) Bilkis invests Tk. 2,400 in the XYZ Bank at 5%. How much additional money must she invest at 8% so that the total annual income will be equal to 6% of her entire investment?

Solution

Let X is the additional amount of money invested at 8%.

So, the entire investment will be $(2400+X)$

According to the question,

Or, 5% of 2400 + 8% of X = 6% of $(2400+X)$ Or, $120 + 0.08X = 144 + 0.06X$

Or, $0.02X = 24$ Or, $2X = 2400$ Or, $X = 1200$

Answer: Tk 12

Shajalal Islami Bank Ltd
Trainee Officer (Cash)

- 1) On selling 17 balls at Rs. 720, there is a loss equal to the cost price of 5 balls.
The cost price of a ball is:

Solution

Let the cost price of a ball is Rs.x

Given, on selling 17 balls at Rs. 720,

there is a loss equal to the cost price of 5 balls

The equation is : $17x - 720 = 5x$ Or, $17x - 5x = 720$ Or, $12x = 720$

or, $x = 60$

Therefore, cost price of a ball is Rs. 60.

Answer: 60

- 2) Salaries of Ravi and Sumit are in the ratio 2:3. If the salary of each is increased by Rs. 4000, the new ratio becomes 40:57. What is Sumit's salary?

Solution: Let, Salary of Ravi is $2x$ & Sumit is $3x$

According to the question,

$$2x + 4000 \quad 3x + 4000 = 40 \quad 57 \quad \text{Or, } 114x + 228000 = 120x + 160000$$

$$\text{Or, } 6X = 68000$$

$$\text{Or, } X = 11,333.33$$

$$\text{Sumit's Salary} = 3 \times 11333.33 = 34000.$$

Answer: 34000

- 3) Two students appeared at an examination. One of them secured 9 marks more than the other and his marks were 56% of the sum of their marks. What are the marks obtained by them?

Solution: Let, One gets x & other One gets $(x + 9)$.

According to the question,

$$X + 9 = (x + x + 9) \times 14 \quad 25 \quad [14 \quad 25 = 0.56] \quad \text{Or, } 25x + 225 = 28x + 126$$

$$\text{Or, } 3x = 99 \quad \text{Or, } X = 33$$

$$\text{Other one gets } (x + 9) = 33 + 9 = 42$$

Answer: 42

- 4) A man purchased a cow for Rs. 3000 and sold it the same day for Rs. 3600, allowing the buyer a credit of 2 years. If the rate of interest be 10% per annum, then the man has a gain of:

Solution

Here, Principle = Tk. 3000,

interest rate = 10%,

time = 2years

$$\text{We know that, } I = Pnr = 3000 \times 2 \times 10\% = 3000 \times 2 \times \frac{10}{100} = 600$$

$$\text{So, the future value} = 3000 + 600 = 3600$$

Selling price of the cow is Rs. 3600.

$$\text{Therefore, the profit or gain by selling the cow is} = 3600 - 3600 \quad 3600 \times 100 = 0\%$$

The man has a gain of 0% as the cost price and selling price are the same

Alternative way

20% CP: FV = 5: 6 = 3000: 3600 FV = 3600 SP = 3600 No Loss No Gain
 Answer No Loss No Gain

- 5) A father told his son, “I was as old as you are at present at the time of your birth”. If the father’s age is 38 years now, the son’s age five years back was?

Solution

Let the son's present age be x years.

According to the question,

Difference of father's and son's present age = Son's present age (38 - x) = x

Or, 2x = 38 Or, x = 19

Five years back = (19 - 5) = 14

Answer 14 years

Bangladesh Bank
Assistant Director

1)
$$\frac{4\sqrt{p} \times 16^{pq} \times 2^{-2}\sqrt{q}}{8\sqrt{q} \times 4^{pq} \times 2^{-3}\sqrt{q}} \times \frac{4^{pq} \times 2^{4pq} \times 2^{-2}\sqrt{p}}{2^3\sqrt{q} \times 2^{2pq} \times 2^{-3}\sqrt{q}} = \frac{2^2\sqrt{p} \times 2^{4pq} \times 2^{-2}\sqrt{p}}{2^3\sqrt{q} \times 2^{2pq} \times 2^{-3}\sqrt{q}} \times 2^{-2pq}$$

Solution

$$\frac{4\sqrt{p} \times 16^{pq} \times 2^{-2}\sqrt{q}}{8\sqrt{q} \times 4^{pq} \times 2^{-3}\sqrt{q}} \times 4^{pq} = \frac{2^{2\sqrt{p}} \times 2^{4pq} \times 2^{-2}\sqrt{p}}{2^{3\sqrt{q}} \times 2^{2pq} \times 2^{-3}\sqrt{q}} \times 2^{-2pq}$$

$$= 2^{2\sqrt{p} + 4pq - 2\sqrt{p} - 3\sqrt{q} + 2pq - 3\sqrt{q}} \times 2^{-2pq} = 2^{4pq - 2pq} \times 2^{-2pq} = 2^{2pq - 2pq} = 2^0 = 1 [\text{Answer}]$$

- 2) Every year Mr Kalam saves Tk. 5400 which is 15% of her annual income. Mr Rahim spends Tk. 2800 a month which is 87.5% of her monthly income. Who has more income?

Solution

Let, Mr Kalam’s annual income = Tk x 15% × x = 5400 Or, x = 5400 / 15 × 100 = 36000

His monthly income = Tk 36000/ 12 = Tk 3000

Let, Mr Rahim’s income = Tk y 87.5% × y = 2800

Or, y = 2800 / 87.5 × 100 = 3200

Mr Rahim has monthly income [Answer]

- 3) The average weight of A, B and C is 40kg. Weight of C is 24 kg more than A's weight and 3 kg less than B's weight. What will be the average weight of A, B, C and D, if D weights 15 kg less than C?

Solution

Here, $A + B + C = 120$

Or, $A + B + C = 120$

Given, $C - A = 24$

Or, $C = A + 24$ ----- (1)

Again, $B - C = 3$

Or, $B = 3 + C$

Or, $B = A + 24 + 3$

Or, $B = A + 27$ ----- (2)

Now, $A + B + C = 120$

Or, $A + A + 27 + A + 24 = 120$

Or, $3A + 51 = 120$

Or, $A = 69 / 3 = 23$

From eq (1),

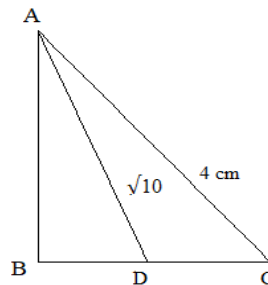
$C = 23 + 24 = 47$

From eq (2), $B = 47 + 3 = 50$

Given $C - D = 15$ Or, $D = C - 15 = 47 - 15 = 32$

Hence, average weight of A, B, C & D is $(A + B + C + D) / 4 = (23 + 50 + 47 + 32) / 4 = 152 / 4 = 38$ kg [Answer]

- 4) In the figure below, AB is perpendicular to BC and $BD = DC$, If $AD = \sqrt{10}$ cm and $AC = 4$ cm, then what is the value of BC?



From the figure, $AD = \sqrt{10}$, $AC = 4$ cm. Let $BD = DC = x$ cm and $AB = h$ cm

Here, ABD is a right angle triangle, from the Pythagorean Theorem

$AD^2 = AB^2 + BD^2$ Or,

$(\sqrt{10})^2 = h^2 + x^2$

Or, $h^2 = 10 - x^2$(1)

Again, from triangle ABC

$$AC^2 = AB^2 + BC^2 \quad [BC = BD + DC = x + x = 2x]$$

$$\text{Or, } 4^2 = h^2 + (2x)^2$$

$$\text{Or, } 16 = 10 - x^2 + 4x^2 \quad \text{Or,}$$

$$3x^2 = 16 - 10 = 6 \quad \text{Or, } x^2 =$$

$$2$$

$$\text{Or, } x = \sqrt{2}$$

$$\text{Hence, } BC = x + x = 2\sqrt{2} \text{ cm [Answer]}$$

- 5) **Matin visited his cousin Raju during the summer vacation. In the mornings, they both would go for swimming. In the evenings, they would play tennis. They would engage in at most one activity per day, i.e. either they went swimming or played tennis each day. There were days when they took rest and stayed home all day long. There were 32 mornings when they did nothing, 18 evenings when they stayed at home, and a total of 28 days when they swam or played tennis. What duration of the summer vacation did Matin stay with Raju?**

Solution

Let, total days = d

Matin, went for swimming = (d - 32) days

Matin, played tennis for (d-18)days

According to the question,

$$(d - 32) + (d - 18) = 28$$

$$\text{Or, } 2d = 28 + 50 \quad \text{Or, } d =$$

$$\frac{78}{2}$$

$$2$$

$$\therefore d = 39$$

Matin stayed with raju for 39 days. [Answer]

Bangladesh Bank Officer

- 1) **A labor is paid Tk.100 per hour for working 8 hours in a day as regular pay, His payment is 1.5 times than the rate of regular pay in excess of 8 hours in a single day as overtime. If the labor received a total of Tk.1000 for a single day, how long did he work on that day?**

Solution

Let. A labor does “x” hour extra duty

According to the question

$$100 \times 8 + x \times 100 \times 1.5 = 1000$$

$$\text{Or, } 100(8 + 1.5x) = 1000$$

$$\text{Or, } 8 + 1.5x = \frac{1000}{100} = 10$$

$$\text{Or, } 1.5x = 10 - 8 = 2$$

$$\text{Or, } x = \frac{2}{1.5} = 1.33$$

Total hours he works = $8 + 1.33$ hours

= 9.33 hours = 9 hours 20 mins

Another way

For 8 hours duty a labor gets = Tk 100×8 = Tk 800 Extra

payment he gets = Tk $(1000 - 800)$ = Tk 200 Extra per hour

payment = Tk 100×1.5 = Tk 150 Extra time he works

$$= \frac{200}{150} \text{ hours} = 1.33 \text{ hours}$$

Total hours he works = $8 + 1.33$ hours = 9.33 hours = 9 hours 20 mins

Answer: 9 hours 20 mins

2) Mr. A deposited a certain amount of money for a fixed period of time. On maturity, he received a total of Tk 60,000 and the ratio of interest and investment was 3:6. If the interest rate was 6.25%(simple). Calculate the time period for which the money was invested?

Solution

Ratio method

$$\text{Interest} = \text{Tk } 60000 \times \frac{3}{9} = \text{Tk } 20000$$

$$\text{Investment (Principal)} = \text{Tk } 60000 \times \frac{6}{9} = 40000$$

If principal is 100 then interest is 6.25 in 1 year

$$\text{If principal is 40000 then interest is 20000 in } = \frac{20000 \times 100}{6.25 \times 400} = 8 \text{ years}$$

Formula method

Let, Interest, $I = 3x$, Investment, $P = 6x$

Here, rate of interest, $r = 6.25\%$

We know,

$$I = Pnr$$

$$\text{Or, } 3x = 6x \times n \times 6.25\%$$

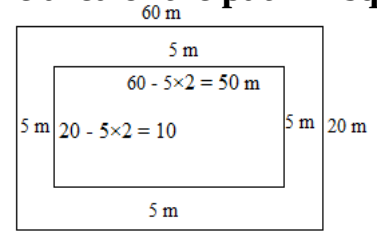
$$\text{Or, } n = \frac{3 \times 100}{6.25 \times 6} = 8\%$$

Answer: 8%

- 3) The length and width of a garden is 60 meter and 20 meter respectively. Inside the garden, there is a 5 meter-wide path around it. What is the area of the path in square meter?

Solution

Area including path = $60 \times 20 = 1200$ sq. m. Length without path = $60 - (5 \times 2) = 50$ m Width without path = $20 - (5 \times 2) = 10$ m Area without path = $50 \times 10 = 500$ sq. m.
Area of the path = $1200 - 500 = 700$ sq. m.



Answer: 700 sq. m

- 4) In an office, $\frac{1}{3}$ rd of the workers are women, $\frac{1}{2}$ of the women are married and $\frac{1}{3}$ rd of the married women have children. If $\frac{3}{4}$ th of the men are married and $\frac{2}{3}$ rd of the married men have children, what parts of workers are without children?

Solution

Let, total number of workers = x
Number of women = $x/3$ part
Number of men = $x - x/3 = 2x/3$ part
Women married = $\frac{1}{2} \times x/3 = x/6$ part
Women having children = $\frac{1}{3} \times x/6 = x/18$ part
Married Men = $\frac{3}{4} \times 2x/3 = x/2$ part
Men having Children = $\frac{2}{3} \times x/2 = x/3$ part
So, Workers with children = $x/3 + x/18 = 7x/18$ part
Workers without children = $x - 7x/18 = 11x/18$ part
So, Part of workers without children = $11x/18 / x = 11/18$ part

- 6) 200 children came to park with bicycle and tricycle All the older children used bicycles and all the younger children used tricycles for coming to the park. As such there were 480 wheels altogether that rode into the park. Find the number of elder children and younger children?

Solution

Let, number of elder children = x and number of younger children = $200 - x$

According to the question

$$2x + 3(200 - x) = 480 \text{ Or, } 2x + 600 - 3x = 480$$

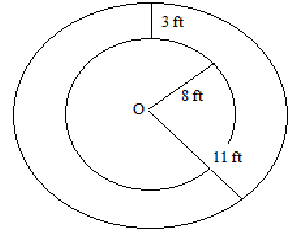
$$\text{Or, } x = 600 - 480 = 120$$

Therefore, number of elder children = 120 and number of younger children = $200 - 120 = 80$

Answer: 120 and 80

- 1) The figure shows a circular flowerbed, with its center at O having radius of 8 feet. The flowerbed is surrounded by a circular path, which is 3 feet wide. What is the area of the path in square feet?

Solution



The radius of the bigger circle is $8 + 3 = 11$ feet, thus its area is $= \pi r^2 = \pi \times 11^2 = 121\pi$ sq feet

The radius of smaller circle is 8 feet.

The area of the smaller circle is $= \pi r^2 = \pi \times 8^2 = 64\pi$ sq feet

The difference is $= (121 - 64)\pi = 57\pi$ sq feet

Answer 57π sq feet

- 2) A trader bought some mangoes for Tk. 150 per dozen and an equal number of apples for Tk. 100 per dozen. If he sells all the fruits for Tk. 140 per dozen, what be his profit/loss in percentage?

Solution Given

that,

Selling Price of 1 dozen fruits is, SP = Tk. 140

Cost Price of 1 dozen fruits is, CP = Tk. $\frac{150 + 100}{2} = \text{Tk. } 125$

Profit of 1 dozen fruits is = Tk. $(140 - 125) = 15$ tk

Now, Profit% $= \frac{15}{125} \times 100\% = 12\%$

Answer 12%

- 3) Three friends invest equal shares in a business. After careful calculation, they realized that each of them would have invested Tk. 3,000 less if they had found 2 more friends with equal shares. What is the amount of total investment in the business?

Solution

Let, each of the three friends invested Tk. x and total investment of three friends = Tk.

$3x$ According to the question,

$3x = 5(x - 3000)$ Or, $3x =$

$5x - 15000$ Or, $2x = 15000$

So, $x = 7500$

Therefore, total investment, $3x = \text{Tk. } 3 \times 7500 = \text{Tk. } 22500$

Answer Tk. 22500

4) Taxi fare is described by the following relationship:

Total taxi Fare = Tk. A upto 2 km + Tk. B per km (exceeding 2 km) + Tk.60 for per hour Waiting.

A person paid Tk.432 for running 52 km and 2 hours of waiting charge. The same person paid Tk.732 for running 102 km and 2 hours of waiting charge. Find the value of "A" and "B".

Solution

Here, Total taxi Fare = Tk. A upto 2 km + Tk. B per km (exceeding 2 km) + Tk.60 for per hour waiting

Equation	Distance		Waiting time	Total Tk.
first equation	k. 2A	k.50B	k. 60×2	32
econd equation	k. 2A	k. 100B	k. 60×2	32

Therefore, $2A + 50B = 432 - 120 = 312$. (i)

And $2A + 100B = 732 - 120 = 612$. (ii)

(ii) – (i),

$2A - 2A + 100B - 50B = 612 - 312$ Or, $50B = 300$

So, $B = 6$

Putting $B = 6$ at any of the equations,

$2A = 312 - 50 \times 6 = 12$

Or, $A = 6$

Answer A = 6 and B = 6

1. In a rectangular field length is $\frac{6}{5}$ of breadth, if the perimeter is 132, then the area of the field?

Let, length = $6x$
Breadth = $5x$
 $2 \times (5x + 6x) = 132$
 $\Rightarrow x = 6$
Length = 36, Breadth = 30
Area = $36 \times 30 = 1080$ sq.unit

2. Sum of present age of A and B is 60; after 5 years the ratio of their age 3:4, present age of A is?

Let, A = x
B = $60 - x$
 $(x + 5) / (60 - x + 5) = 3/4$
 $\Rightarrow x = 25$

3. A:B=2:3 & B:C=5:6, Amount is 430 then B's share?

A:B=2:3 $\times 5 = 10:15$
B:C=5:6 $\times 3 = 15:18$
(10+15+18)=43 for 430
So 15 for 150

4. When a product is sold for Tk 700 the gain is thrice the amount when it sold for Tk 500 as loss. Cost price of the product?

Let, Cp = x
 $3 \times (x - 500) = 700 - x$
 $\Rightarrow x = 550$ Tk. Ans.

5. When a student got 38% marks failed by 6 marks, when got 45% marks he got 15 marks more than the passing marks. What's the percentage of passing?

Let, total marks x
 $38\% x + 6 = 45\% x - 15$
 $\Rightarrow 7\% x = 21$
 $\Rightarrow 100\% x = 300$
Passing mark = $300 \times 38\% + 6 = 120$
Required % = $120/300$ or 40% Ans.

Bangladesh Bank AD 2023 (Based 2021)

প্রশ্নঃ ৪২ সে.মি. ব্যাসার্ধের বৃত্তাকার তারকে বাকিয়ে একটি আয়ত তৈরী করা হলো। আয়তের বাহুগুলোর অনুপাত ৬ : ৫ হলে, আয়তের ছোট বাহুর দৈর্ঘ্য কত?

সমাধানঃ

দেয়া আছে, বৃত্তাকার তারের ব্যাসার্ধ, $r = ৪২$ সেমি

$$\therefore \text{বৃত্তাকার তারের পরিধি} = ২\pi r = ২ \times \frac{২২}{৭} \times ৪২ = ২৬৪ \text{ সেমি}$$

ধরি, আয়তক্ষেত্রে বাহু দুইটি যথাক্রমে $৬x$ এবং $৫x$

$$\text{আয়তক্ষেত্রের পরিসীমা} = ২(৬x + ৫x) = ২ \times ১১x = ২২x$$

$$\text{প্রশ্নমতে, } ২২x = ২৬৪$$

$$\Rightarrow x = \frac{২৬৪}{২২} = ১২ \text{ সেমি}$$

$$\therefore \text{ছোট বাহুর দৈর্ঘ্য} = ৫x = ৫ \times ১২ = ৬০ \text{ সেমি।}$$

প্রশ্নঃ সোনা ও তামামিশ্রিত ১২ গ্রাম গহনায় সোনা ও তামার অনুপাত ২:১। ১৮ গ্রাম ভরের অনুপাত ৫:১ হলে সম্পূর্ণ মিশ্রণের অনুপাত কত?

সমাধানঃ ১২ গ্রাম মিশ্রণে,

$$\text{সোনার পরিমাণ} = \frac{২}{২+১} \times ১২ = ৮ \text{ গ্রাম}$$

$$\text{তামার পরিমাণ} = \frac{১}{৫+১} \times ১২ = ৪ \text{ গ্রাম}$$

১৮ গ্রাম মিশ্রণে,

$$\text{সোনার পরিমাণ} = \frac{৫}{৫+১} \times ১৮ = ১৫ \text{ গ্রাম}$$

$$\text{তামার পরিমাণ} = \frac{১}{৫+১} \times ১৮ = ৩ \text{ গ্রাম}$$

সম্পূর্ণ মিশ্রণে,

$$\text{সোনার পরিমাণ} = ৮ + ১৫ = ২৩ \text{ গ্রাম}$$

$$\text{তামার পরিমাণ} = ৪ + ৩ = ৭ \text{ গ্রাম}$$

নির্ণেয় অনুপাত

$$\text{সোনা : তামা} = ২৩ : ৭$$

প্রশ্নঃ একটি ট্রেন ২ সেকেন্ডে দাঁড়িয়ে থাকা ব্যক্তিকে অতিক্রম করে। ৪.৫ সেকেন্ডে ৫০মি. দৈর্ঘ্যের একটি প্লাটফর্মকে অতিক্রম করে। ট্রেনের দৈর্ঘ্য ও গতিবেগ কত?

সমাধানঃ

ধরি, ট্রেনটির দৈর্ঘ্য = x মি. এবং গতিবেগ = y মি./সে.

দেয়া আছে, ট্রেনটি ২ সেকেন্ডে দাঁড়িয়ে থাকা ব্যক্তিকে অতিক্রম করে।

প্রশ্নমতে, $x = 2y$

আবার, $50 + x = \frac{9}{2}y$

$\rightarrow 50 + 2y = \frac{9}{2}y$

$\rightarrow 8y = 100 + 8y$

$\rightarrow 8y - 8y = 100 \Rightarrow 4y = 100 \Rightarrow y = 20$

ট্রেনের গতিবেগ = ২০ মি./সে.

ট্রেনের দৈর্ঘ্য = $2y = 2 \times 20 = 40$ মি.

প্রশ্নঃ কোনো আসল ৩ বছরে চক্রবৃদ্ধি ৬৬৯০ টাকা এবং ৬ বছরে ১০০৩৫ টাকা হলে আসল কত? সমাধানঃ

ধরি, আসল, $P = x$ টাকা

দেয়া আছে, সময়, $n = 3$ বছর এবং মুনাফা-আসল, $A = 6690$ টাকা।

আমরা জানি, $A = P \left(1 + \frac{r}{100}\right)^n$

$\Rightarrow 6690 = x \left(1 + \frac{r}{100}\right)^3 \dots\dots\dots(1)$

আবার, সময়, $n = 6$ বছর এবং মুনাফা-আসল, $A = 10035$ টাকা।

আমরা জানি, $A = P \left(1 + \frac{r}{100}\right)^n \Rightarrow 10035 = x \left(1 + \frac{r}{100}\right)^6 \dots\dots\dots(2)$

সমীকরণ (২) কে সমীকরণ (১) দ্বারা ভাগ করে পাই, $= \frac{x \left(1 + \frac{r}{100}\right)^6}{x \left(1 + \frac{r}{100}\right)^3} = \frac{10035}{6690}$

$\Rightarrow \left(1 + \frac{r}{100}\right)^3 = \frac{10}{9} \dots\dots\dots(3)$

সমীকরণ (৩) এর মান সমীকরণ (১) বসিয়ে পাই,

$\Rightarrow 6690 = x \times \frac{10}{9} \Rightarrow x = 6690 \times \frac{9}{10} \Rightarrow x = 6021$ টাকা।

Combined Senior Officer 2021 Based

Exam Date: 08/11/2023

- 1) There are three containers A, B and C which contain water, milk and acid respectively in equal quantities. 10% of the content of A is taken out and poured into B. Then, the same amount from B is transferred to C, from which again the same amount is transferred to A. What is the proportion of milk in container A at the end of the process?

Solution:

Let Each container has 100 liters of water, milk and acid respectively.

1st cases,

As 10% of 100 = 10 liter has taken out and poured in B.

So, Total Quantity in Container B = 100 + 10 = 110 liter

Now, Milk Concentration in B = $\frac{100}{110} = \frac{10}{11}$ Liter

2nd cases,

Again, 10 liter of B is poured to C

So, Milk in Container C = $10 \times \frac{10}{11} = \frac{100}{11}$ liter

Now, Milk Concentration in C = $\frac{100}{11} \div 110 = \frac{10}{121}$ liter

3rd cases,

Further, 10 liter of C is poured to A

So, Milk in Container A = $10 \times \frac{10}{121} = \frac{100}{121}$ Liter

Now, Milk Concentration in A = $\frac{100}{121} \div 100 = \frac{1}{121}$ Liter

So, Required Proportion of milk in container A at the end of the process $\frac{1}{121}$ Liter (Ans)

2. Rima invests Tk. 5000 in a bond which gives interest at 4% per annum during the first year, 5% during the second year and 10% during the third year. How much does she get at the end of the third year?

In 1st Year, Amount of Interest = $5000 \times \frac{4}{100} = 200$ Tk

In 2nd Year, Amount of Interest = $(5000 + 200) \times \frac{5}{100} = 260$ Tk

In 3rd Year, Amount of Interest = $(5200 + 260) \times \frac{10}{100} = 546$ Tk

So, Rima gets at the end of the third year = $5000 + 200 + 260 + 546 = 6006$ tk (Ans)

3. Ten different letters of an alphabet are given. Words with letters are formed from these given letters. Then what will be the number of words which have at least one letter repeated?

Solution:

Here, Number of words which have at least one letter repeated = Total - No letter is being Repeated.

So, Total = $10^5 = 100000$ and No letter is being Repeated = $10P5 = 10 \times 9 \times 8 \times 7 \times 6 = 30240$

So, Number of words which have at least one letter repeated = 69760 (Ans)

4. Given : $\frac{2}{x} + \frac{2}{3y} = \frac{1}{6}$ and $\frac{3}{x} + \frac{2}{y} = 0$ Find a for which $y = ax - 4$

Solutino: Here, $\frac{2}{x} + \frac{2}{3y} = \frac{1}{6}$ (I) and $\frac{3}{x} + \frac{2}{y} = 0$ (ii)

Now, Multiplying (ii) by $\frac{1}{3}$ we get, $\frac{1}{x} + \frac{2}{3y} = 0$ So, $\frac{2}{3y} = -\frac{1}{x}$(iii)

Again from equation (i) we get , $\frac{2}{x} - \frac{1}{x} = \frac{1}{6}$ so, $x = 6$

Putting $x=6$ in (iii) we get, $\frac{2}{3y} = -\frac{1}{6}$ So, $y = -4$ (iv)

Now, $y = ax - 4$ So, $-4 = a \times 6$ [from (iii) and (iv) So, $a=0$ (Ans)

5. In the given figure $\angle BAD = \angle CAD$. $AB = 4$ cm, $AC = 5.2$ cm, $BD = 3$ cm. Find BC.

According to question, Given that

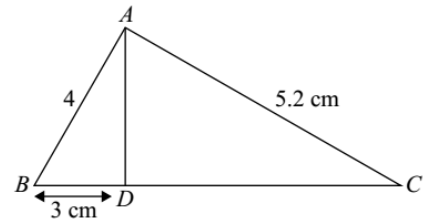
$\angle BAD = \angle CAD$. $AB = 4$ cm, $AC = 5.2$ cm, $BD = 3$ cm

In $\triangle ABC$, AD is the bisector of $\angle A$

$$\frac{AB}{AC} = \frac{BD}{CD} \text{ (Internal bisector prop.)}$$

$$\frac{4}{5.2} = \frac{3}{DC} \Rightarrow DC = 3.9 \text{ cm}$$

But $BC = BD + CD = 3\text{cm} + 3.9 \text{ cm} = 6.9 \text{ cm}$ (Ans)



Janata Bank RC 2020 Based
Exam date: 9/11/2023

1. With reference to a number greater than one, the difference between itself and its reciprocal is 25% of the sum of itself and its reciprocal. By how much percentage is the fourth power of the number greater than its square?

Solution:

Let the number be x.

According to the question:

$$X - 1/x = 25/100 \times (x + 1/x)$$

$$\Rightarrow x^2 - 1 = 1/4 (x^2 + 1)$$

$$\Rightarrow 4x^2 - 4 = x^2 + 1$$

$$\Rightarrow 3x^2 = 5$$

$$\Rightarrow x^2 = 5/3$$

$$\Rightarrow x^4 = 25/9$$

$$\therefore \text{Required percentage} = \frac{(\frac{25}{9} - \frac{5}{3})}{\frac{5}{3}} \times 100 = 66.7\% \text{ (Ans)}$$

2. Find the value of $\sqrt{2+\sqrt{3}} \times \sqrt{2+\sqrt{2+\sqrt{3}}} \times \sqrt{2+\sqrt{2+\sqrt{2+\sqrt{3}}}} \times \sqrt{2-\sqrt{2+\sqrt{2+\sqrt{3}}}}$

Given That, $\sqrt{2+\sqrt{3}} \times \sqrt{2+\sqrt{2+\sqrt{3}}} \times \sqrt{2+\sqrt{2+\sqrt{2+\sqrt{3}}}} \times \sqrt{2-\sqrt{2+\sqrt{2+\sqrt{3}}}}$

$$\text{So, } \sqrt{2+\sqrt{3}} \times \sqrt{2+\sqrt{2+\sqrt{3}}} \times \sqrt{(2)^2 - \left(\sqrt{2+\sqrt{2+\sqrt{3}}}\right)^2}$$

$$\text{So, } \sqrt{2+\sqrt{3}} \times \sqrt{2+\sqrt{2+\sqrt{3}}} \times \sqrt{2-\sqrt{2+\sqrt{3}}}$$

$$\text{So, } \sqrt{2+\sqrt{3}} \times \sqrt{2-\sqrt{3}}$$

$$\text{So, } \sqrt{2^2 - (\sqrt{3})^2} = \sqrt{4-3} = 1$$

So, Required Value of the Given Expression be 1 (Ans)

3. A ship 77 km from the shore, springs a leak which admits 2 tonnes of water in 5 minutes. 92 tonnes of water would sink it. But the pumps can throw out 12 tonnes of water per hour. Find the averages rate of sailing so that the ship may just reach the shore as it begins to sink.

$$\text{Rate of leak} = 2.25 \times \frac{60}{5.5} = \frac{270}{11} \text{ tonnes/hr and rate of pump} = 12 \text{ tonnes/hr}$$

$$\text{To suffice 92 tonnes, time taken by ship} = 92 \div \left(\frac{270}{11} - 12 \right)$$

$$= 92 \times \frac{11}{138} = \frac{22}{3} \text{ hr}$$

$$\text{Average rate of sailing} = \text{distance/time}$$

$$= 77 \div \frac{22}{3}$$

$$= 77 \times \frac{3}{22} = 10.5 \text{ (Ans)}$$

4. In a mixture of milk and water the proportion of milk is 60% by weight. If from 80 gm of mixture, 20gm mixture taken out and 6 gm of water added to the mixture. Then find the ratio of milk and water in the new mixture.

$$\text{Percentage of milk in the mixture} = 60\%$$

$$\therefore \text{Quantity of milk in 80 gm mixture}$$

$$\Rightarrow \text{Quantity of milk} = (60/100) \times 80$$

$$\Rightarrow \text{Quantity of milk} = 48 \text{ gm}$$

$$\text{Quantity of water in the mixture} = 80 - 48 = 32 \text{ gm}$$

$$\text{Now, 20 gm of mixture is removed and 6 gm of water is added}$$

$$\therefore \text{Quantity of milk in mixture} = 48 - (20 \times 48/80) = 48 - 12 = 36$$

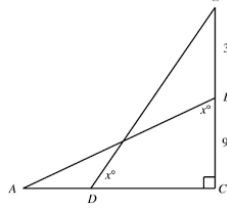
$$\text{Quantity of water in mixture} = 32 - (20 \times 32/80) + 6 = 38 - 8 = 30$$

$$\text{Ratio of milk and water in the new mixture}$$

$$\Rightarrow \text{Ratio} = 36/30 = 6/5 = 6 : 5$$

$$\therefore \text{Ratio of milk and water in the new mixture is } 6 : 5 \text{ (Ans)}$$

5. In the figure, what is the area of $\triangle ABC$ if $\frac{EC}{CD} = 3$?



Solution:

From the given figure, $EC = 9 + 3 = 12$

Given that, $EC/CD = 3$. So, $CD = 4$

But the triangle ABC and EDC are similar.

So, $EC/AC = CD/BC \Rightarrow 12/AC = 4/9 \Rightarrow AC = 27$

Finally we get, base 27 and height 9 for the triangle ABC.

So, the area of the triangle ABC = $0.5 \times 27 \times 9 = 121.5$ (Ans)